

Exploring the Effects of Real-Time Machine Translation and Transcription on Non-Native Speakers During Online Interviews

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Abstract

In recent years, real-time machine translation (MT) and transcription technologies have been widely utilized for multilingual online communication. Previous research has focused primarily on group meetings, exploring the effects of MT tools on non-native speakers' (NNS) comprehension or native speakers' (NS) communication strategies. However, limited attention has been paid to the impact of translation and transcription on NNS in interview settings, particularly from a psychological perspective. In this study, we conducted a 2 (translation vs. no translation) $\times$ 2 (transcription vs. no transcription) between-subjects experiment involving 80 NNS interviewees and 4 NS interviewers. The results revealed that: (1) translation alone significantly enhanced NNSs' perceived self-efficacy, clarity and responsiveness. (2) Combining translation and transcription led to lower ratings of self-efficacy, clarity and responsiveness, compared to using either one alone. (3) Neither tool had a significant effect on NNSs' perceived comfort or workload. These findings suggest that translation captions offer the most effective support for NNSs in online interviews, and highlight the importance of aligning communication tools with the nature of multilingual interactions. This study provides practical implications for designing inclusive and linguistically supportive video conferencing environments.

CCS Concepts

• **Human-centered computing** $\rightarrow$ **Empirical studies in HCI**.

Keywords

Multilingual Communication; Video Conferencing; Real-time Machine Translation; Transcription

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1 Introduction

In recent years, video conferencing (VC) has become a common medium for remote communication, with built-in platform features that can influence interaction dynamics [23]. While traditional in-person communication remains valuable, it often leads to communication challenges and scheduling difficulties [33]. As a result, online interviews have gained popularity and are widely used in settings such as journalistic interviews with eyewitnesses or key informants (e.g., [40]) in news reporting, doctor–patient consultations (e.g., [43]), and academic interview studies (e.g., [35]). However, online interviews may be particularly challenging for non-native speakers (NNS) due to reduced access to nonverbal cues [38], technical disruptions [15], and asymmetries (e.g., [20, 29]) that can affect their ability to understand and respond effectively.

To address the challenges NNS face in online communication, investigators have developed machine translation (MT) and artificial intelligence (AI) tools to support them (e.g., [9, 22]). Although these tools have enhanced multilingual interactions, most assessments focused on meeting contexts rather than interview settings, with limited research on how they affect NNSs' perceptions in online interviews. Unlike group meetings, interviews are more asymmetric and fast-paced, placing greater pressure [32] and cognitive load on NNS. In addition to communication challenges, online interviews may also create psychological burdens for NNS, including heightened anxiety (e.g., [31]) and reduced self-efficacy (e.g., [16]), which can affect their performance [20] and perception. These factors make it essential to study how MT and AI tools affect NNS perceptions in online interviews. Therefore, we proposed the following question:

RQ: How will transcription and translation affect the perception of NNS during online interviews?

We begin by reviewing the literature on online interviews, multilingual communication, machine translation tools, and video conferencing. Following this, we describe our study design, which adopts a 2 $\times$ 2 between-subjects experimental framework. The findings show that either transcription or translation alone improves NNSs' self-efficacy, clarity, and responsiveness, while combining

both reduces these benefits. Neither tool had a significant effect on perceived workload or comfort. By examining the impact of these tools in online interview settings, our study provides insights for designing real-time language support systems that better meet the needs of NNSs in online interviews.

2 Related Work

This section discussed the advantages and limitations of online interviews for NNS, with particular attention to their challenges and the potential of transcription and translation tools to provide support in asymmetric multilingual contexts.

2.1 Advantages, Constraints and Asymmetries of Online Interviews

With technological advancements over the last few decades, online interviews have addressed many of the limitations associated with onsite interviews (e.g., [24]), particularly since the outbreak of the COVID-19 pandemic. Previous studies have identified many advantages of online interviews, such as the video technology that enables researchers to observe research settings remotely. While group interviews have their strengths, scheduling in-person meetings can be difficult [7], which is an issue that video technology helps to overcome.

Despite these advantages, online interviews pose challenges for NNS. Compared to face-to-face settings, online interviews limit access to nonverbal cues like head tilt, gestures, and facial expressions [38], which are particularly important for individuals from high-context cultures [26]. The restricted screen view in online interviews further reduces the visibility of these cues, potentially increasing misunderstanding for NNS. In addition, technical disruptions such as jitter [15] may further strain NNS, who already face a higher cognitive load when communicating in a second language [40], leaving fewer mental resources to manage disruptions. Moreover, interview settings are often asymmetric, especially in English-speaking contexts where interviewers are usually NS or highly fluent NNS. This puts NNS at a linguistic (e.g., [20]) and interactional disadvantage (e.g., [29]).

Unlike team meetings that involve multiple participants and encourage balanced contributions (e.g., [14]), one-on-one interviews are inherently asymmetric, with the interviewer asking and the interviewee responding. Delays in understanding or replying can be judged negatively [32], not only in high-stakes contexts (e.g., [41]), but also in public settings like media interviews. Thus, the pressure to respond quickly increases NNSs' cognitive burden [11]. To support communication in such settings, many tools have been developed for multilingual online interviews (e.g., [4, 49]).

2.2 Linguistic and Psychological Challenges for NNS in Multilingual Communication

In the global context, multilingual communication is increasingly needed across various domains (e.g., [5]). However, linguistic, cultural, and logistical barriers still hinder effective global communication (e.g., [34, 44, 47]). For NNS, these challenges are amplified. Accents can create barriers and lead to misunderstandings [39], and NS may unconsciously speak too fast [46], increasing processing pressure. NNS often avoid asking for clarification due to fear of

embarrassment or disturbing others [10], making comprehension even harder. Furthermore, foreign language anxiety can cause NNS to feel nervous or avoid interactions altogether [48].

In this context, the challenges faced by NNS are not only linguistic barriers but also psychological burdens, particularly heightened language anxiety and stress when communicating in English [31]. Many NNS also blame themselves for communication breakdowns, which can harm impression management [20]. In light of these psychological burdens, it is essential to explore whether machine translation tools in multilingual communication can offer not only psychological but also communicative benefits to NNS. This perspective also underscores a key motivation for our current study.

2.3 The Potential of Transcription and Translation in Asymmetric Online Interviews

Building on the earlier discussion, this study investigates whether real-time transcription and translation can help compensate for technological limitations and enhance communication both functionally and psychologically.

Previous research has examined automatic speech recognition (ASR) transcripts as mediation tools in audio conferencing (e.g., [19, 36]). While ASR generates transcripts in the original language, machine translation (MT) tools have also been used to support multilingual communication through dual translations [13], tailored strategies [49], and visual enhancements like text highlighting [12]. However, these tools have mostly been studied in symmetric, text-based, or task-oriented group settings. Some earlier studies reported that MT could introduce confusion due to translation errors [41], though recent advances in neural MT have significantly improved translation quality [8]. With real-time MT becoming increasingly available in video conferencing platforms (e.g., [17, 21]), it is worthwhile to explore how these newer MT tools function in online interview contexts, which differ from group communication in structure and interaction roles.

In summary, while prior work has demonstrated the value of ASR and MT tools in multilingual collaboration, few studies have examined how these tools function in asymmetric one-on-one interactions. This study extends existing research by investigating the psychological and communicative effects of real-time transcription and translation on NNS in online interview settings.

3 The Current Study

In this study, we implemented real-time English captions and Chinese translations in online one-on-one interviews, with NS as interviewers and NNS as interviewees. There were four different conditions for NNS interviewees: no support, transcription only, translation only, and both transcription and translation.

Based on the above argument, we drew on the concept of self-efficacy to investigate the psychological effects of MT on NNS. Self-efficacy refers to individuals' beliefs in their ability to meet specific performance standards, which in turn shapes their thoughts, behaviors, and emotional responses [1, 2]. It has also been linked to English proficiency among NNS teachers [16]. As we used transcription and translation to support NNS in understanding their partners in the interview, they might believe themselves to have the

capabilities to perform better. Therefore, we proposed the following hypotheses and the research question:

H1a: NNS will rate their self-efficacy higher with transcription than without it.

H1b: NNS will rate their self-efficacy higher with translation than without it.

RQ1: How will transcription and translation together affect the self-efficacy of NNS?

Previous studies have pointed out that real-time transcription can help NNS better understand the communication content and interact with it (e.g., [37]). Perceived communication quality includes three keys: Clarity refers to the degree of comprehension of the meaning being communicated. Responsiveness indicates the norm of coordination or reciprocity that individuals experience in interpersonal interactions. Comfort is defined as a condition of positive effect of ease and pleasantness when interacting with each other [30]. Based on this, we proposed the following:

H2a: NNS will rate their perceived quality of communication higher with transcription than without it.

H2b: NNS will rate their perceived quality of communication with translation than without it.

RQ2: How will transcription and translation together affect the perceived quality of communication of NNS?

In addition, people often need to invest extra time and effort in reading transcriptions or translations during online interviews. This additional burden could potentially offset the advantages we propose. To explore this further, we asked the following question:

RQ3: How will transcription and translation affect the workload of NNS?

4 Method

We designed a 2 (transcription: present or absent) x 2 (translation: present or absent) between-subjects experiment to explore the effects of real-time translation and transcription on NNS interviewee experiences. Participants were assigned to one of four conditions: none, translation only, transcription only, or both. Each participant were interviewed by a native English-speaking trained confederate. Afterwards, they completed an online survey about the self-efficacy, perceived quality of communication, and workload.

4.1 Participants

We recruited 80 native Chinese speaking participants (54 female, 26 male, mean age 22.56 years) from a large U.S. university. They had lived in the United States for an average of 4.09 years ($SD = 3.66$). Participants' self-rated English proficiency was moderate ($M = 5.35$, $SD = 0.88$), and they had minimal experience using translation or transcription tools ($M = 3.59$, $SD = 1.74$). Participants were randomly assigned to one of four conditions: no support, transcription only, translation only, or both transcription and translation ($N = 20$ per condition).

4.2 Interview and Equipment

We provided a structured interview guide that included 21 questions on aspects of American culture. The questions covered three main topics: food, sports, and campus life. Each topic contained six basic questions, two specific items, and one open-ended question. During the interviews, they were instructed to ask questions at a steady pace and to not deviate from the script. Interviews were conducted via Zoom using built-in, real-time closed captioning for both transcription and translation, with captions displayed at the bottom of the video feed during conversations (see Figure 1).



Figure 1: The participant's Zoom interview interface.

4.3 Procedure

Before joining the Zoom meeting, the interviewer was renamed "Chris" and the interviewee was renamed "Alex" (gender neutral names) to protect participant privacy. Participants first signed a digital consent form. Then, they read the instructions for interviewees. The interviewer then asked the questions following the provided outline, and the interviewee responded accordingly. None of the participants or confederates had access to Zoom's additional features during the experiment. The average interview duration was 25 minutes and 43 seconds. After completing the task, participants completed an online post-task survey about their experiences.

4.4 Measures

Three constructs were assessed at the individual level: self-efficacy, perceived communication quality (clarity, responsiveness, comfort) and workload.

Self-efficacy. The participants reported their self-efficacy on a 7-point scale [6]. The 8 questions formed a reliable scale (Cronbach's $\alpha = .94$) and were averaged to create a measure of "Self-efficacy".

Perceived Quality of Communication. To assess how participants experienced the quality of communication during the interview, we asked them to complete three rating sub-scales after the online interview. Each scale used a 7-point Likert format and was adapted from a measure originally developed by Liu et al. [30]. The instrument included three dimensions: clarity (Cronbach's $\alpha = .83$), responsiveness (Cronbach's $\alpha = .86$), and comfort (Cronbach's $\alpha = .92$).

Workload. We assessed perceived workload using six 7-point items adapted from the NASA TLX [18]. Five items formed a reliable scale (Cronbach's $\alpha = .79$) and were averaged to create a measure of "Workload".

5 Results

To examine the effects of machine translations and transcriptions on NNS, we conducted a series of 2 (transcription) $\times$ 2 (translation) ANOVAs with self-efficacy, perceived quality of communication and workload as dependent variables.

5.1 Self-Efficacy

Our H1a, H1b, and RQ1 explored the perceived self-efficacy of participants during online interviews. The results of the 2 (transcription) $\times$ 2 (translation) ANOVA revealed no significant main effects. Participants' self-efficacy did not differ significantly across conditions, whether with transcription captions ($M = 5.34$, $SE = 0.16$) vs. without ($M = 5.23$, $SE = 0.16$), $F(1, 76) = 0.24$, $p = .62$, or with translations ($M = 5.47$, $SE = 0.16$) vs. without ($M = 5.10$, $SE = 0.16$), $F(1, 76) = 2.62$, $p = .11$.

However, the ANOVA test showed a significant interaction between transcription and translation on the interviewees' self-efficacy: $F[1, 76] = 6.79$, $p = .01$ (see Figure 2). Pairwise comparisons showed that participants in the translation-only condition ($M = 5.71$, $SE = 0.29$) and the transcription-only condition ($M = 5.45$, $SE = 0.29$) reported significantly higher self-efficacy than those in condition with no support ($M = 4.74$, $SE = 0.29$; for translation-only, $p < .001$; for transcription only, $p = .03$). The translation-only condition yielding the highest mean score. The combined-support condition ($M = 5.23$, $SE = 0.29$) also showed relatively high self-efficacy compared to the no-support condition, though it was lower than either single-support condition.

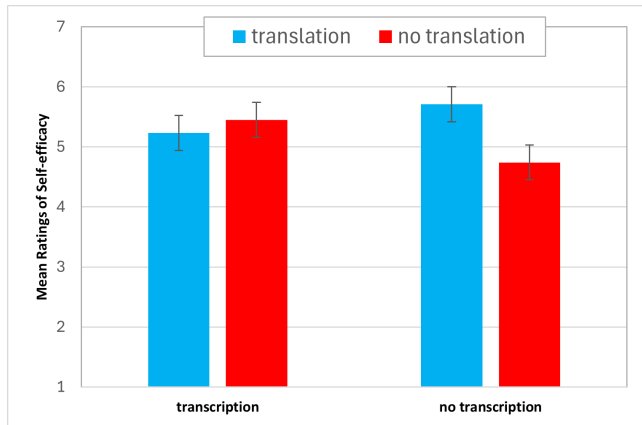


Figure 2: Mean self-efficacy on a scale of 1 (low) to 7 (high), by condition. Error bars represent the standard errors of the mean.

5.2 Perceived Quality of Communication

H2a, H2b, and RQ2 focused on participants' perceived quality of communication during online interviews. We assessed communication quality using three subscales: clarity, responsiveness, and comfort. Each one was analyzed through separate ANOVAs. Comfort showed no significant effects, whereas clarity and responsiveness revealed significant interactions in response to transcription and translation support.

No significant main effects of transcription ($F[1, 76] = 0.42$, $p = .52$) or translations ($F[1, 76] = 0.59$, $p = .45$) were found on clarity, but their interaction was significant: $F[1, 76] = 4.93$, $p = .03$ (see Figure 3). Pairwise comparisons revealed that translation-only condition reported the highest perceived clarity ($M = 5.96$, $SE = 0.25$), significantly higher than no-support ($M = 5.22$, $SE = 0.25$; $p = .038$) and both supports ($M = 5.25$, $SE = 0.25$; $p = .046$). The transcription-only condition ($M = 5.61$, $SE = 0.25$) also showed higher perceived clarity than the no-support and combined-support conditions.

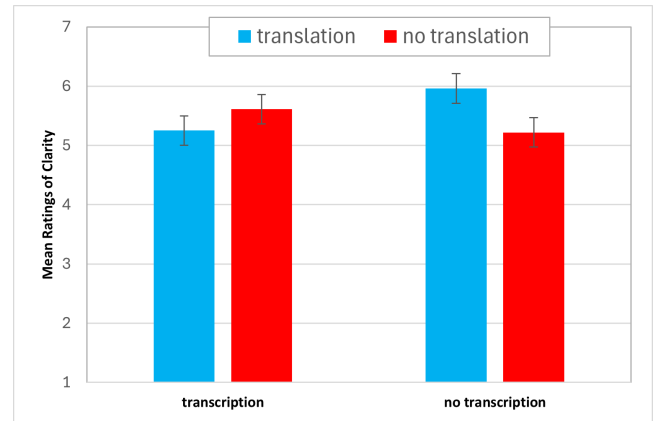


Figure 3: Mean ratings of clarity on a scale of 1 (low) to 7 (high), by condition. Error bars represent the standard errors of the mean.

Neither transcription ($F[1, 76] = 0.30$, $p = .58$) nor translation ($F[1, 76] = 0.19$, $p = .66$) had a significant main effect on responsiveness, but their interaction was significant: $F[1, 76] = 8.21$, $p = .005$ (see Figure 4). We also ran pairwise comparisons and translation-only condition again reported the highest perceived responsiveness ($M = 6.34$, $SE = 0.23$), which was significantly higher than no-support condition ($M = 5.59$, $SE = 0.23$; $p = .02$) and both support condition ($M = 5.56$, $SE = 0.23$; $p = .02$). The transcription-only condition also showed relatively high responsiveness ($M = 6.11$, $SE = 0.23$).

5.3 Workload

To answer RQ3, we conducted a 2 (transcription) $\times$ 2 (translation) ANOVA using our workload scale as a dependent measure. The results showed no significant main effect of transcription ($F[1, 76] = 0.05$, $p = .82$), translation ($F[1, 76] = 2.23$, $p = .14$) or their interaction effects ($F[1, 76] = 2.63$, $p = .11$).

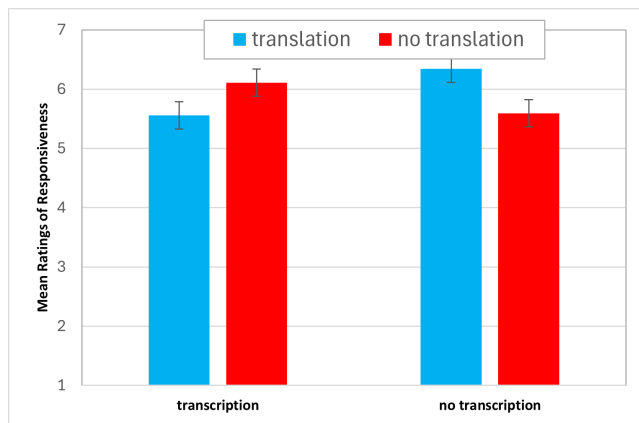


Figure 4: Mean ratings of responsiveness on a scale of 1 (low) to 7 (high), by condition. Error bars represent the standard errors of the mean.

6 Discussion

In this discussion, we highlight two key findings: first, providing either transcription or translation alone can facilitate more transparent communication and boost confidence among NNSs; second, using both simultaneously may introduce new challenges that hinder interaction quality in online interviews.

Participants who received translation reported significantly higher levels of self-efficacy, clarity and responsiveness compared to those in the no-support or combined-support conditions. The transcription-only condition also reported significantly higher self-efficacy than the non-support condition. These results suggest that using either tool individually can enhance NNSs' experience of real-time communication. Among the three measures, self-efficacy stands out, underscoring the core contribution of our study: these tools not only improve communication quality but also boost NNSs' confidence and sense of control in asymmetric interactions. Moreover, the translation condition consistently produced the highest scores, likely because it enabled native-language processing and reduced cognitive load [28]. In contrast, transcription still required second-language processing, though it may have supported comprehension by allowing participants to re-read spoken content, especially when unfamiliar vocabulary or accents were present (e.g., [9, 45]). Overall, even simple tools, when used individually, can meaningfully enhance both communication and psychological empowerment for NNSs.

Although each tool showed benefits on its own, the combined condition ranked third in self-efficacy and clarity, lowest in responsiveness, and was significantly worse than translation alone in both clarity and responsiveness. One explanation is that errors from machine-generated transcription and translation may have compounded, reducing effectiveness. In addition, switching between alphabetic language (e.g., English) and logographic language (e.g., Chinese) may have disrupted comprehension due to structural differences [25], and slowed responses due to cognitive switching costs [3]. This suggests that combining language supports requires

careful integration to avoid overwhelming users in real-time communication.

Finally, contrary to our hypotheses, neither transcription nor translation significantly affected workload or comfort. Comfort may have been shaped more by interpersonal and contextual factors (e.g., [42]), and closed captions may have supported cognitive processing (e.g., [27]), keeping workload stable.

There are several limitations in this study. First, we focused solely on the effects of transcriptions and translations on NNSs' self-efficacy, perceived quality of communication, and workload. More psychological dimensions should be investigated in the future. Besides, the findings of this study are based on self-reported data from participants. Future research could examine participants' spoken messages to objectively assess the impact of our tools. Moreover, conducting in-depth interviews with a subset of participants could provide additional insights.

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